

CABLEFLOW-AI

Stochastic Forecasting Framework for XLPE Demand Optimization

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Project Repository:

<https://github.com/salmawaleed055/OptiLayer-Forecaster>

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Abstract

CableFlow-AI introduces a specialized intelligence layer for XLPE material management. By synchronizing 25 years of regional macroeconomic volatility, industrial tonnage requirements, and global price indices, the system provides a mathematically validated roadmap for proactive procurement in Egypt, UAE, and Bahrain markets.

1 Data Acquisition & Processing

The framework utilizes a high-dimensional dataset synchronized to a monthly frequency (2000–2025). The following features were processed and expanded:

- **xlpe_demand_Million_tons:** Annual historical tonnage expanded to monthly intervals using **Cubic Spline Interpolation**. Missing values were estimated using a **Geometric Growth approach**.
- **polyethylene_price:** Raw material price indices expanded from annual benchmarks to monthly intervals using **Cubic Spline Interpolation**.
- **Average Inflation index of Dollar:** Global inflation trends expanded from annual reported indices using **Cubic Spline Interpolation**.
- **gdp_growth_rate:** Annual data (2000–2024) expanded via **Cubic Spline Interpolation**. Missing 2025 data generated using a **Monte Carlo Bootstrap** method.
- **Total electricity consumption (ME):** Extracted via *WebPlotDigitizer*. Data was smoothed and a **stochastic variation ($\pm 1\%$)** was introduced.

1.1 Expansion Methodology

The transition from discrete annual points to a continuous monthly time-series was achieved through two

primary statistical layers:

1. Geometric Continuity (Cubic Spline):

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

2. Stochastic Expansion (2024–2025):

$$X_{m+1} = X_m + \epsilon, \quad \epsilon \sim \mathcal{N}(\mu_m, \sigma_m)$$

2 Data Analysis & Feature Engineering

Dataset summary: 312 records (28/02/2001 – 31/12/2025).

Lag Features: lag_1, lag_3, lag_12, rolling_mean_3.

Infrastructure Proxy Features: construction_output_index (Mean 109.69), urbanization_rate (Mean 54.09%), infrastructure_investment (Mean 0.266329M tons).

2.1 Predictor Features Statistics

- polyethylene_price: Mean 8371.88, Std 869.99
- Average Inflation index of Dollar: Mean 2.95, Std 1.95
- gdp_growth_rate: Mean 3.56, Std 2.36
- Total electricity consumption, ME: Mean 1,014,008.53, Std 364,296.63
- Monthly growth rate, ME %: Mean 0.50, Std 2.72
- construction_output_index: Mean 108.89, Std 5.90
- urbanization_rate: Mean 55.29, Std 19.86
- infrastructure_investment: Mean 0.2677, Std 0.0298
- Lag features and rolling means also included

3 Methodology

CableFlow-AI follows a specialized two-stage pipeline:

1. **Predictive Forecasting:** An ensemble-based learning architecture predicts regional XLPE demand using:
 - Linear Regression (Enhanced)
 - Ridge Regression
 - Random Forest (Risk-Aware)
 - Gradient Boosting (Risk-Aware)

2. **Supply Chain Optimization:** Dynamic Inventory Engine based on EOQ (Economic Order Quantity):

$$EOQ = \sqrt{\frac{2DS}{H}}$$

3.1 Overfitting Awareness

Random Forest and Gradient Boosting show higher training accuracy but overfit historical data. Ensemble modeling reduces variance and improves generalization for unseen periods (2026–2027).

3.2 Optimizations Explained

- **Linear Regression:** Weighted lags, asymmetric loss, risk-adjusted for price trends.
- **Random Forest:** Tree-level uncertainty, dynamic safety stock, price-based hedging.
- **Gradient Boosting:** Staged predictions, sequential learning, uncertainty tracking.

3.3 Time-Series Forecasting Framework

Objective: Predict XLPE demand using historical patterns and macroeconomic indicators.

Practices: Chronological train/test split, TimeSeriesSplit CV (Cross-Validation), lag-based features, pipeline preprocessing, GridSearchCV for hyperparameters, ensemble VotingRegressor.

4 Model Performance Summary

Model KPIs

Linear Regression (Enhanced): Accuracy 97.84%, MAE 0.006448M tons, R^2 0.444, Overfitting 0.488
Ridge Regression: Accuracy 98.39%, MAE 0.004742M tons, R^2 0.6822, Overfitting 0.362
Random Forest (Risk-Aware): Accuracy 98.64%, MAE 0.004144M tons, R^2 0.7061, Overfitting 0.2869
Gradient Boosting (Risk-Aware): Accuracy 97.72%, MAE 0.007101M tons, R^2 0.5036, Overfitting 0.4944
Ensemble (Top 3): Accuracy 98.60%, MAE 0.004187M tons, R^2 0.7347, Overfitting 0.2462

5 Inventory Optimization Engine

This module translates AI forecasts into practical procurement decisions using classical inventory theory. The system takes dashboard inputs and generates optimal stock policies.

5.1 Core Functions

- **Safety Stock:** Uses AI forecast uncertainty σ with service level z -score

$$SS = z \times \sigma \times \sqrt{LT}$$

- **Economic Order Quantity (EOQ):** Minimizes total inventory cost

$$EOQ = \sqrt{\frac{2DS}{H}}$$

- **Reorder Point:** When to place new orders

$$ROP = \text{Lead Time Demand} + SS$$

5.2 Dashboard Inputs (User Configurable)

Parameter	Value
Material Cost	\$2,350/ton
Holding Cost	\$75/ton/month
Ordering Cost	\$7,500/order
Stockout Penalty	\$1,200/ton
Service Level	95%
Lead Time	30 days

5.3 Code Architecture

- **Input:** AI forecast CSV + user parameters
- **Process:** Safety stock → EOQ → Reorder point → Cost analysis
- **Output:** JSON recommendations + CSV summaries
- **Features:** What-if scenarios, risk assessment, cost optimization

5.4 Sample Output

- Safety Stock: 0.0039M tons
- EOQ: 0.1287M tons
- Reorder Point: 0.0654M tons
- Annual Cost: \$36.8M
- Service Level: 95%

The optimizer balances AI predictions with real-world costs to recommend when and how much to order.

6 Infrastructure-Based Scenario Forecasting

6.1 Scenario-Based Forecasting for XLPE Cable Demand

To estimate XLPE cable requirements for future infrastructure (8,779.9 km²), we used a scenario-based approach. Since historical infrastructure data is unavailable, we calculated the cable demand from planned projects and integrated it with the model's forecast.

Step 1 – Planned Infrastructure Assessment: All planned infrastructure projects in the Middle East were collected, and their covered areas were recorded. These areas were then used to estimate the total amount of XLPE cables required using the formula:

$$\begin{aligned} \text{Total Cable Mass (tons)} \approx & \text{Infrastructure Area (km}^2\text{)} \\ & \times \text{Cable Density (km/km}^2\text{)} \\ & \times \text{Cable Weight (t/km)} \end{aligned} \quad (1)$$

- Infrastructure Area: 8,779.9 km²
- Cable Density: 100 km/km²
- XLPE Cable Weight: 1 t/km

$$\text{Total Cable Mass} \approx 8,779.9 \times 100 \times 1 \approx 878,000 \text{ tons}$$

Step 2 – Integration with Forecast Model: The infrastructure-based cable mass was added directly to the forecasted XLPE demand from the time-series model. This approach treats planned infrastructure as an additional scenario input rather than a historical feature.

Step 3 – Benefits:

- Enables realistic XLPE demand projections even without historical infrastructure data.
- Supports scenario analysis and procurement planning.
- Clearly separates market-driven demand (from model) and infrastructure-driven demand (from planned projects).

6.2 2026 Scenario Forecast

- Model-Predicted XLPE Demand: 0.389 million tons
- Infrastructure Contribution: 0.878 million tons
- Total Projected Demand: 1.267 million tons
- Infrastructure represents 69.3% of total demand
- Market-driven component: 30.7%

6.3 Integration with Ensemble

- Lower variance (70% reduction)
- RF 60%, Ridge 40% combination
- Safety Stock (1.96σ): 0.003892M tons (1.27%)

7 Novelty

- Multi-variable macroeconomic synchronization
- Infrastructure-to-Tonnage Calculation Engine
- Risk-aware ensemble intelligence
- Scenario-based forecasting with predictive hedging

Key Performance Indicators

Forecast Accuracy (1-MAPE): 98.60%
Infrastructure Projected Demand: 1.266848M Tons
Safety Stock Savings (Est): \$18M – \$73M/year

8 Limitations and Data Constraints

This work is subject to several limitations that stem primarily from data availability constraints rather than methodological choices:

- **Use of proxy data for demand:** Cable consumption in the Middle East was used as a proxy for XLPE cable demand. While this reflects overall market trends, it does not isolate demand specific to XLPE insulation.
- **Lack of cable specification details:** Available consumption data is aggregated and does not differentiate between cable sizes, voltage levels, or cross-sectional areas. As a result, the model does not forecast demand for XLPE cables of specific technical specifications.
- **Estimated infrastructure-driven demand:** The calculation of cable demand from future infrastructure projects relies on average cable area, assumed cable lengths, and estimated material density. In practice, accurate estimation requires detailed engineering data such as cable type, cross-sectional area, and insulation thickness.
- **Partial coverage of infrastructure projects:** Future infrastructure data was collected from publicly announced projects in major Middle Eastern countries. Not all countries or projects could be captured due to limited publicly available information, potentially leading to underestimation of total regional demand.
- **Use of proxy variables for key drivers:** Several independent variables (e.g., polyethylene prices) were used as proxies for underlying drivers such

as XLPE material costs. Access to material-specific pricing and utility-level procurement data would improve model robustness.

- **Limited dataset size:** The final dataset contains approximately 300 observations, which limits the learning capacity of the machine learning models compared to large-scale industrial datasets.
- **Synthetic temporal expansion:** Due to the lack of monthly historical data, annual observations were statistically expanded into monthly values. While necessary for time-series modeling, this reduces the precision of short-term and seasonal demand patterns.

These limitations are largely outside the scope of the project. With access to higher-resolution, material-specific, and project-level datasets, the proposed framework would yield more accurate and granular demand forecasts.

9 Conclusion

CableFlow-AI transforms XLPE procurement into a predictive, risk-aware decision system. By integrating infrastructure scenarios with stochastic ensemble modeling and addressing overfitting, it provides a production-ready solution for the MENA cable market.

10 References

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